Factors Influencing Student Performance in Mathematics

Hongjie Deng, Anan Wang, Yi Wang, Ranran Zhang, Jingyi Wu

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1. Abstract

Using the UCI Student Performance dataset, this report analyzes the main elements that influence the student's final mathematics grade (G3).(Cortez,2008) Exploratory analysis and model comparisons lead to the final linear model,with prior academic grades being the foremost predictor. Companion logistic model results confirmed that initial academic performance and supportive family contexts are the chief determinant of the final outcome.

2. Introduction

In 2006, around 40% of young people in Portugal left school early—twice the average of the European Union—showing alarming trends in student achievement and engagement.(Cortez & Silva, 2008) This study baseline attempts to understand what really impacts the academic achievement of students in key subjects.Through the study of the learning environment, family and personal factors, the study attempts to understand what impacts students' academic performance to the highest degree. This should give actionable intelligence to educators and policymakers aimed at performance improvement and dropout rate reduction.In the report, all the data are sourced from Portuguese high schools.

3. Dataset

The student performance dataset was collected from students in two Portuguese secondary schools. Mathematics dataset was used rather than the Portuguese language dataset, which is less influenced by culture and language. This dataset contains 395 students and 33 variables, with no missing variables mentioned in official declaration.

Key variables are three stage grades G1, G2, and G3 (0–20), which are interconnected. We noticed that there exists a large proportion of students (about 40 students) who got 0 in the final grade (G3); absence is a right-skewed factor that relates negatively to G3. Most students got 0 failures as shown in the histogram of failures. Family-related variables such as famrel (quality of family relationships) and parental education show overall high values, which indicates their families are generally well-educated. Social and behavioral factors such as higher (whether the student intends to pursue higher education), Mjob (mother's job), romantic (whether the student is in a romantic relationship), goout (frequency of going out with friends, 1–5), Medu (mother's education level, 0–4), Walc (weekend alcohol consumption) These characteristics, alongside the academic variables such as G2 (second period grade), provide useful context for understanding student performance patterns. These variables provide a solid demographic and social foundation to study factors affecting academic performance in mathematics.

4. Analysis

4.1. Model1. We built our origin model using all selected variables via backwards, forwards stepwise selections, correlation test and related literal, which outputs the following formula:

$$G3 \sim higher + Mjob + romantic + G2 + G1 + Medu + Fedu +$$
$$goout + failures + age + absences + Walc + famrel + activities$$

The coefficient showed that G1 and G2 are the most significant predictors of final grade(G3), indicating early academic performance has a strong effect on G3. However, when checking the linear regression assumptions via residual plots and Q-Q plot, both plots revealed non-normality and heteroscedasticity.

4.2. Model2. To address assumption failures, we performed various transformations on different combinations of variables, including square-root and logarithmic transformation. We then compared different transformation models via cross validation and adjusted R square, and found that square-root transformation to the dependent variable (G3) produced the best performance of the model, improving residual normality and variance stability, with a slight decrease in adjusted R square.

4.3. Model3. Correlation tests between variables revealed a strong linear relationship between G1 and G2 (r=0.85). This indicates that multicollinearity issue exists, which could lead to unstable coefficients, and reduced model reliability. We then used the average value of G1 and G2 to represent the overall early academic performance of students' to eliminate multicollinearity. This transformation successfully improved the coefficient estimates and the adjusted R2 of about 0.62.

Then to achieve the balance between model simplicity and accuracy, we conducted another backwards and forwards stepwise selections on Model3, and got a similar summary from both stepwise selections. After that, we compared all models using 10-fold cross-validation. According to adjusted R square, RMSE and MAE, we finally selected Model3, achieving slightly lower in-sample performance than the full model but demonstrated stronger out-of-sample performance.From the summary, we found that our final model explained about 0.61 variation on G3. Overall, Model 3 provided the optimal balance between accuracy, interpretability, and robustness.

$$\sqrt{G3} = 1.18327 - 0.18168\, romantic_{yes} + 0.23519\, \frac{(G1+G2)}{2} -$$
$$0.10156\, failures - 0.06913\, age + 0.02275\, absences$$
$$+0.07794\, Walc + 0.07788\, famrel - 0.10968\, activities_{yes} + \varepsilon$$

4.4. Logistic model. We noticed that our linear model failed to meet the linear regression assumptions, which motivated us to complement our linear model with a logistic model. We also considered the extreme lower-tail observations of G3. we chose the cutpoint 10 to split the continuous final grade G3 into a binary variable this mid-scale threshold aligns with the natural grading scheme and produces approximately balanced class distributions. Cross-validation also supported 10 as the best split in terms of overall RMSE/MAE. We then convert the observations whose G3 <= 10 to 0 and observations whose G3 > 10 to 1, and fit a logistic regression, with predictors selected from our final linear model. From the logistic model summary, we found that prior grade and strongly significant, while weekend alcohol consumption and famrel are also significant at the 5% level; Other behavioral or social variables show less significant impact. The remaining predictors are not significant according to p-value. This highlights that prior performance and family relationship quality are strongly associated with passing, whereas greater age is associated with a substantially lower probability of passing.These predictors provide complementary evidence to support early inferences system development.

5. Result

Our final linear regression model with transformation on dependent(G3), and average (G1+G2)/2 achieves p-value smaller than 0.05 and adjusted R square 0.62, which explains about 62% of the variation in students' G3(G3). At 5% significant level, (G1+G2)/2, absences , Walc , famrel , romanticyes, age ,failures shows significance for predicting G3, Interpreting the coefficient of the final linear regression model, square root of final grade(G3) is expected to increase by 0.24, when increasing the average of G1 and G2 by 1 level. The corresponding p-value of this combination of predictors is much smaller than 0.05, which indicates the earlier grade did perform well when predicting G3. Holding other factors constant, each additional year of age slightly reduces square root of G3 by -0.06, and being in a romantic relationship lowers it by 0.17. One level increase in famrel increases the square root of G3 by 0.08, while parental education does not significantly affect the square root of G3.

When evaluating the assumptions of our final linear regression model, we can't find any support document or official declaration to justify the observations in mathematics student performance dataset are all independent. Thus, independence assumption does not hold without other justification. From normal Q-Q plot and residual plot, we observed major curvature and unequal variance, implying large deviations from linearity, normality and homoscedasticity. We believe this is affected by a large number of observations whose G3=0.

To preserve data integrity as much as possible with consideration of assumption failures, a logistic model was fitted. Observed from summary, the average of early grades, absences, Walc, famrel, age remain significant in the logistic model. The average earlier grade has 1.87 log-odds,(with 95% CI 10.5-3.33). This implies a one-level increase raises the log-odds of G3 > 10 by 1.87(with Odd Ratio increased by 6.49). famrel shows 1.84 log-odds, which indicates that an on-level increase raises the log-odds of G3>10 by 1.84; Walc has 0.95 log-odds. These patterns confirm that earlier grades and good family relationships indeed drive success, while older age slightly reduces the chance of passing.

6. Conclusion and discussion

Our study shows that students who perform well in the earlier academic performance are more likely to achieve high G3. Besides this reliable factor, strong family support and higher parental education add extra marks on G3. While older age and being in a romantic relationship have slightly small negative effects on G3. Similar predictors between final linear regression model and logistic model indicate earlier good academic performance and family supportive family are key factors to final academic success.

6.1. Limitations.

1. Both models used only academic and demographic information. This has omitted other motivational factors, such as teacher interaction, students' confidence, and school activities, which possibly contribute to predicting students' G3.

2. G3 is unevenly distributed with many zeros, which breaks homoscedasticity, although we have performed a logistic model to complement our result. Another suitable model is needed.

3. After transformations and diagnostics, several assumptions (linearity, homoscedasticity, normality) are still violated.The independence assumption may be violated, as a similar study or living environment is allowed when collecting the data,which has violated the independence assumption.

4. Because the final linear model uses ((G1+G2)/2) to predict final grade, practical deployment often needs to wait for the second-period grade (G2), which may not be practical

In practice, the model can be used by teachers or the education department to identify the students who might get failed in the final mathematics assessment. By inputting students' early grades(G1 and G2), family-relationship level and age, our logistic model can estimate the probability of achieving G3 > 10 academic performance. The academic help and targeted tutoring can then be provided to corresponding students before it is too late.

6.2. Future research:. A refined linear model can be built using only observations with (G3>10) to investigate variation among students who may get more than 10 in G3. Furthermore, we can collect more variables from different angles, including emotion and motivation, to get a well-rounded model. Apart from these, we need to consider non-linear and interaction factors, which might be more suitable for predicting G3. The random forest can also be used to predict, as this does not require too many assumption restrictions. To avoid waiting for (G2), an early-stage model only using G1 could be deployed first to get an earlier warning for the students who might get failed in the G3.

7. Appendix

7.1. Figures.

Figure 1. Correlation among G1,G2 and G3

```
          G1        G2        G3
G1 1.0000000 0.8521181 0.8014679
G2 0.8521181 1.0000000 0.9048680
G3 0.8014679 0.9048680 1.0000000
```

Figure 2. Model3 Summary

```
Call:
lm(formula = sqrt(G3) ~ romantic + I((G1 + G2)/2) + failures +
    age + absences + Walc + famrel + activities, data = math_final_grade)

Residuals:
    Min      1Q  Median      3Q     Max
-2.5470 -0.1854  0.1530  0.3903  1.5758

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.18327    0.50184   2.358  0.01888 *
romanticyes     -0.18168    0.07437  -2.443  0.01502 *
I((G1 + G2)/2)  0.23519    0.01081  21.752  < 2e-16 ***
failures        -0.10156    0.05090  -1.995  0.04673 *
age             -0.06913    0.02848  -2.427  0.01568 *
absences         0.02275    0.00440   5.169 3.78e-07 ***
Walc             0.07794    0.02730   2.854  0.00454 **
famrel           0.07788    0.03859   2.018  0.04428 *
activitiesyes   -0.10968    0.06874  -1.596  0.11140
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6769 on 386 degrees of freedom
Multiple R-squared:  0.6244,    Adjusted R-squared:  0.6166
F-statistic: 80.21 on 8 and 386 DF,  p-value: < 2.2e-16
```

Figure 3. Cross Validation

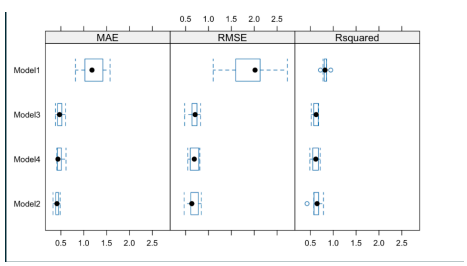

Figure 4. Residual Plot and Normal QQ-plot

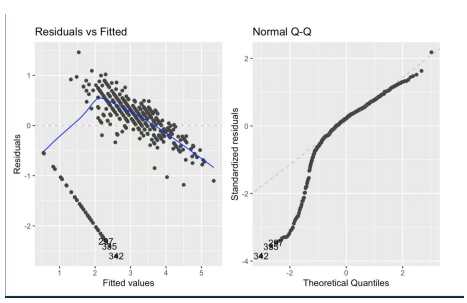

Figure 5. Summary of Logistic Regression Model

	pass		
Predictors	*Log-Odds*	*CI*	*p*
(Intercept)	9.13	-4.72 – 25.36	0.219
(G1 + G2)/2	1.87	1.05 – 3.33	**0.001**
absences	12.21	-43.61 – 447.82	0.993
Walc	0.95	0.14 – 1.99	**0.037**
famrel	1.84	0.64 – 3.52	**0.010**
romantic [yes]	-1.00	-3.04 – 0.79	0.289
activities [yes]	-0.84	-2.94 – 1.07	0.393
age	-1.95	-3.69 – -0.82	**0.006**
Observations	395		
R^2 Tjur	0.855		

7.2. References.

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